
A Topological Characterization for Graph Neural Networks (GNN)

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ML Basics

- ❖ **Dataset** : Sample from a Population with some Probability Distribution D .
- ❖ **Hypothesis** : Model Math function to approximate some “actual” function f . (similar to Regression equation), also called a predictor function. We want to determine the h
- ❖ **Hypothesis Class** : A “Class” of Hypothesis functions i.e. a Set of Polynomials of various degrees.

Sample Instances: A set of instances from a Population,

Split into “disjoint” { Training set, Validation Set, Test Set }

- ❖ **Training Set** : Actually, used as a sample for say n different Hypothesis
- ❖ **Test Set** : Once the final Model is determined, this set is used to assess performance
- ❖ **Validation Set** : This set is used for prediction

A procedure called “Cross-Validation” is used. Divide the Validation set itself into n disjoint “folds” almost equal size

Train the model on this set excluding one. Check performance of the model on the excluded one. Note h_i Hypothesis(predictor function). Select Best h_i , Retrain it on Training Set + Validation set , evaluate performance on Test Set.

Epoch - A single such training + test + validation ‘sessions’

- ❖ **No-Free-Lunch Theorem** : “All optimization algorithms perform equally well when their performance is averaged across all possible problems” , Unknown D and Unknown f implies can’t be sure if one Hypothesis will perform better than others in reality.
- ❖ **Universal Approximation Theorem** (for Neural Nets) -A universal approximation theorem formally states that a family of neural network functions is a dense set within a larger space of functions they are intended to approximate.

Universal Approximation Theorem- for Neural Nets

Universal approximation theorem—Let $C(X, \mathbb{R}^m)$ denote the set of **continuous functions** from a subset X of a Euclidean $\mathbb{R}^n$ space to a Euclidean space $\mathbb{R}^m$. Let $\sigma \in C(\mathbb{R}, \mathbb{R})$. Note that $(\sigma \circ x)_i = \sigma(x_i)$, so $\sigma \circ \mathbf{x}$ denotes σ applied to each component of $\mathbf{x}$.

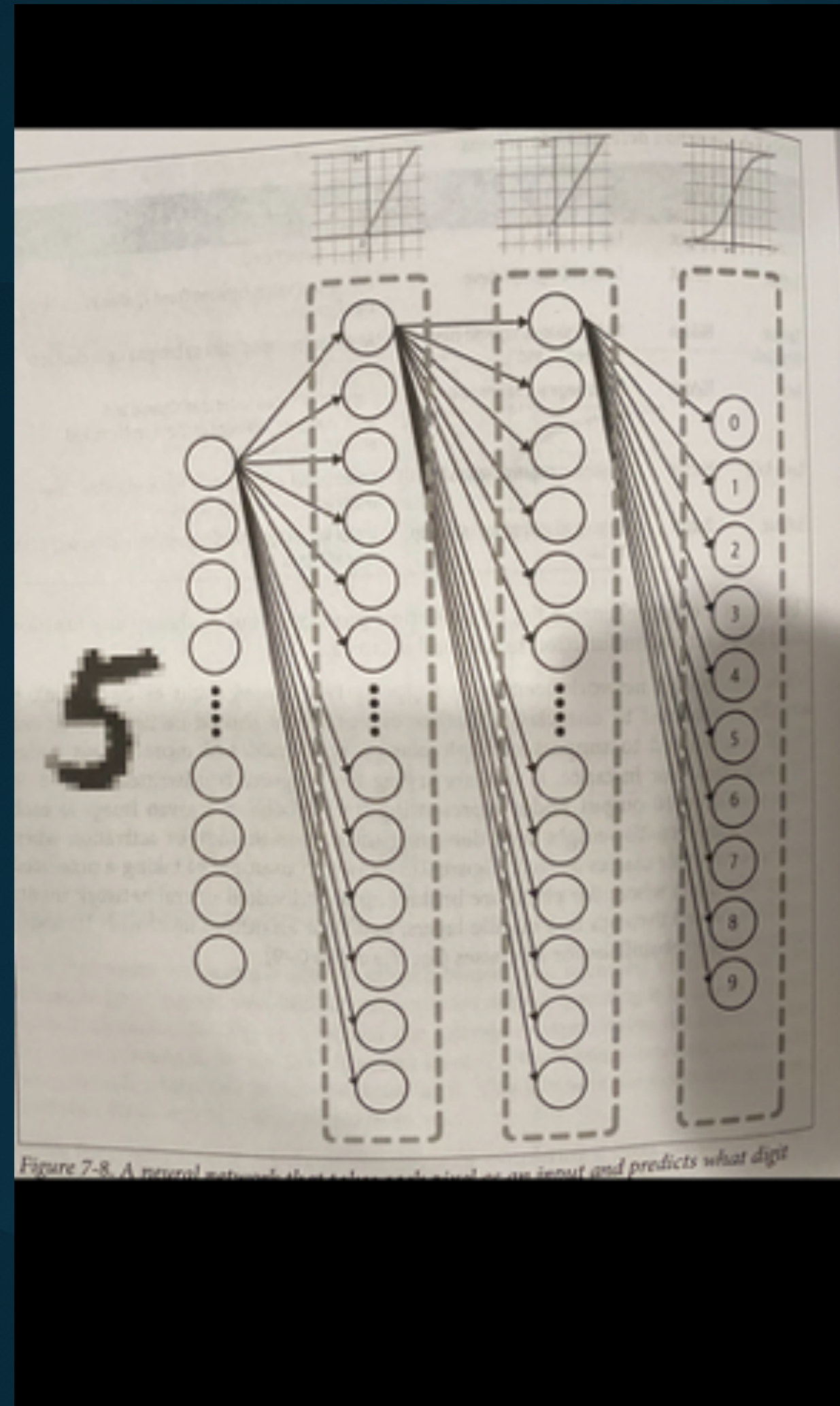
Then σ is not **polynomial if and only if** for every $n \in \mathbb{N}$, $m \in \mathbb{N}$, **compact** $K \subseteq \mathbb{R}^n$, $f \in C(K, \mathbb{R}^m)$, $\varepsilon > 0$ there exist $k \in \mathbb{N}$, $A \in \mathbb{R}^{k \times n}$, $b \in \mathbb{R}^k$, $C \in \mathbb{R}^{m \times k}$ such that

$$\sup_{x \in K} \|f(x) - g(x)\| < \varepsilon$$

where $g(x) = C \cdot (\sigma \circ (A \cdot x + b))$

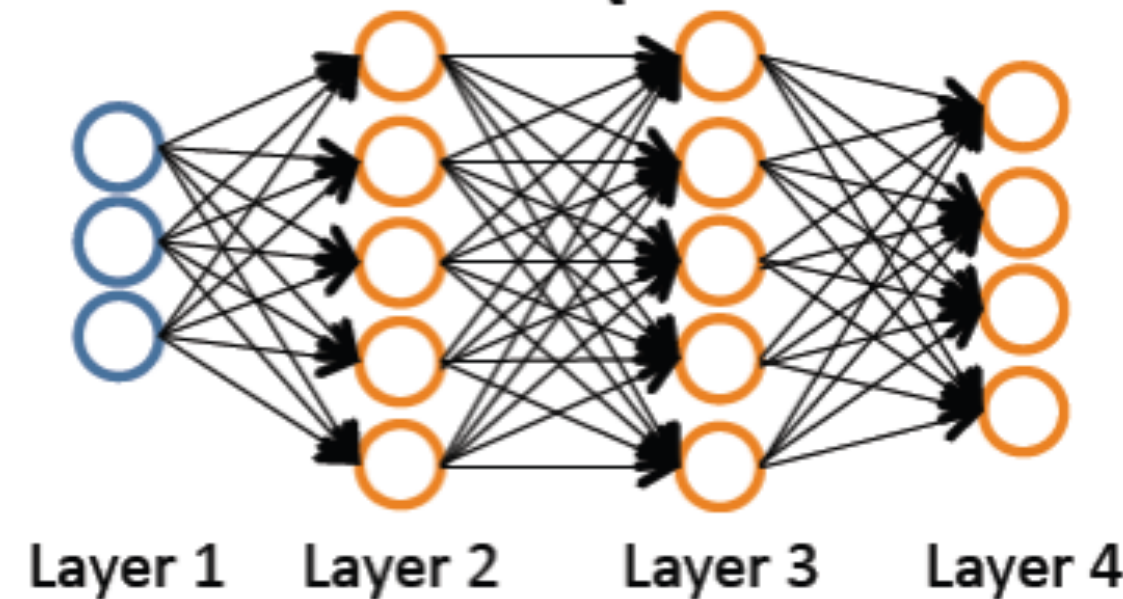
Optical Character Recognition

Example:



Feed forward Neural Nets (Multi-layer Perceptron) – For Non-linear Models

Neural Network (Classification)



Binary classification

$y = 0$ or 1

1 output unit

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$$

$L =$ total no. of layers in network

$s_l =$ no. of units (not counting bias unit) in layer l

Multi-class classification (K classes)

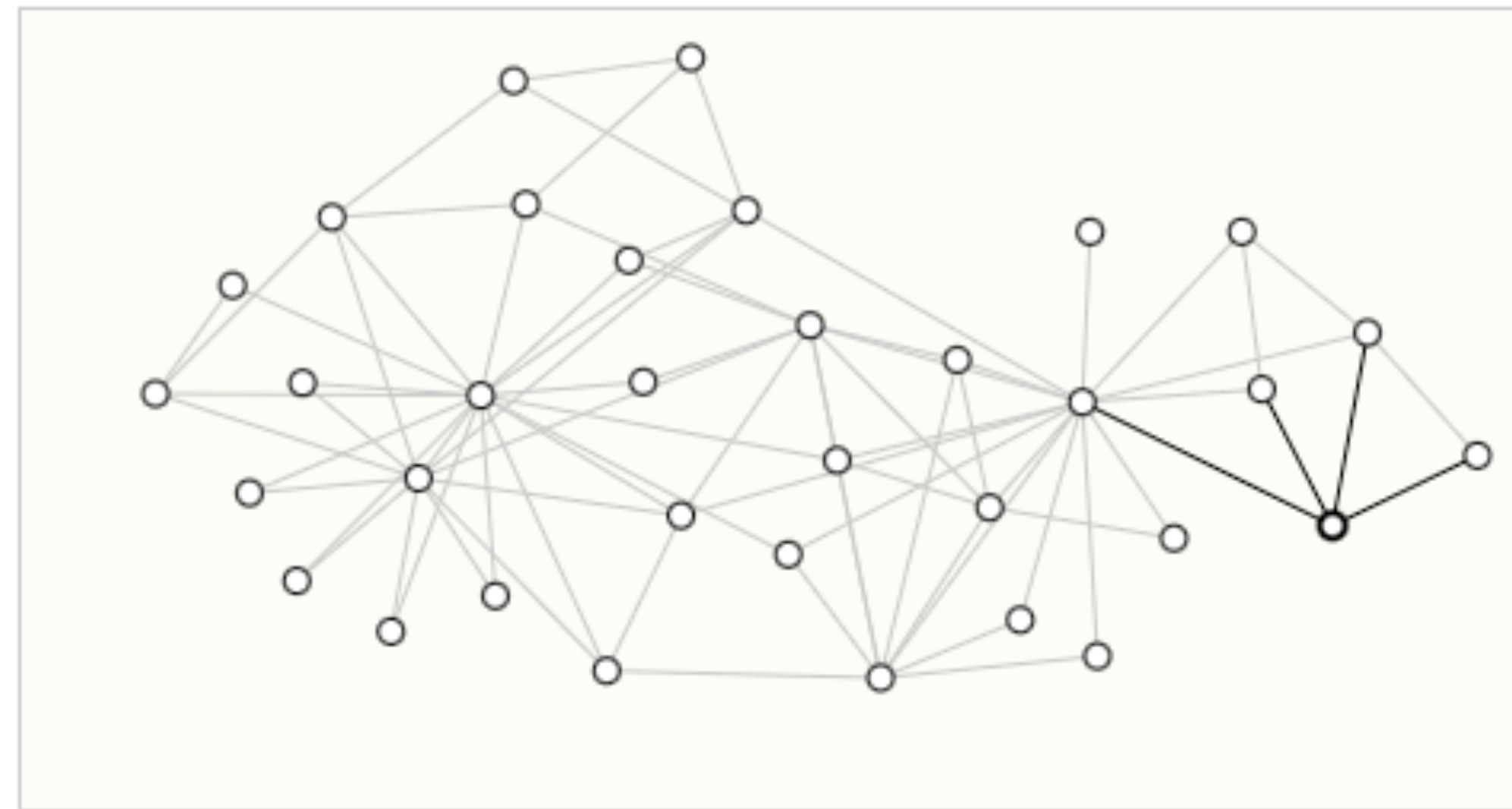
$y \in \mathbb{R}^K$ E.g. $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
pedestrian car motorcycle truck

K output units

ML Pipeline

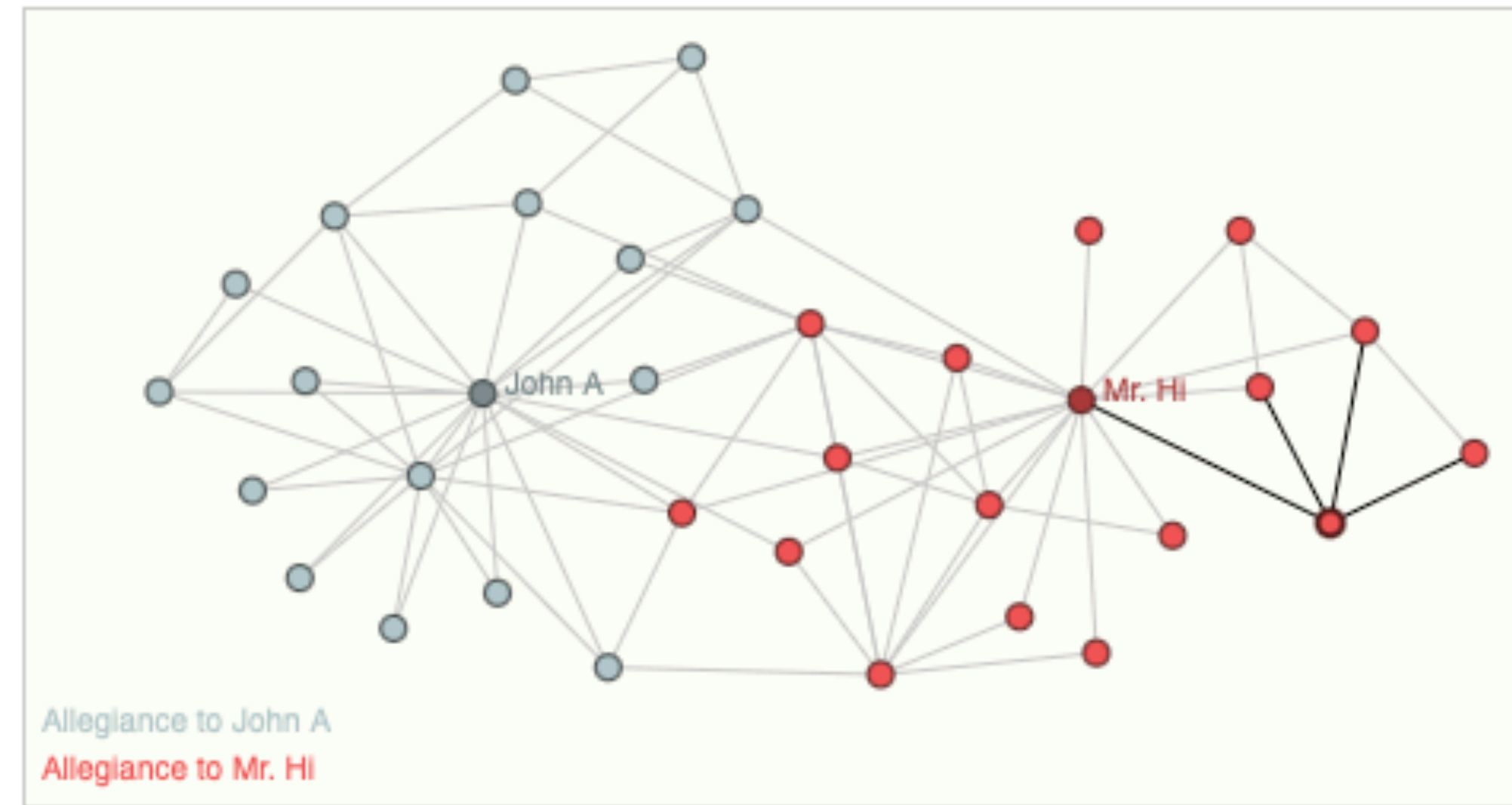
- ❖ Split dataset into Training, Validation, Test
- ❖ Choose a Hypothesis class(or several)
- ❖ Define a Metric of Error i.e. Loss function
- ❖ Use Cross-Validation to find best suited hypothesis class H among selected ones
- ❖ Once a Hypothesis class is selected, train a prediction function h in H and evaluate performance on Test to check “Generalization Error”

Graph Neural Nets(GNN)



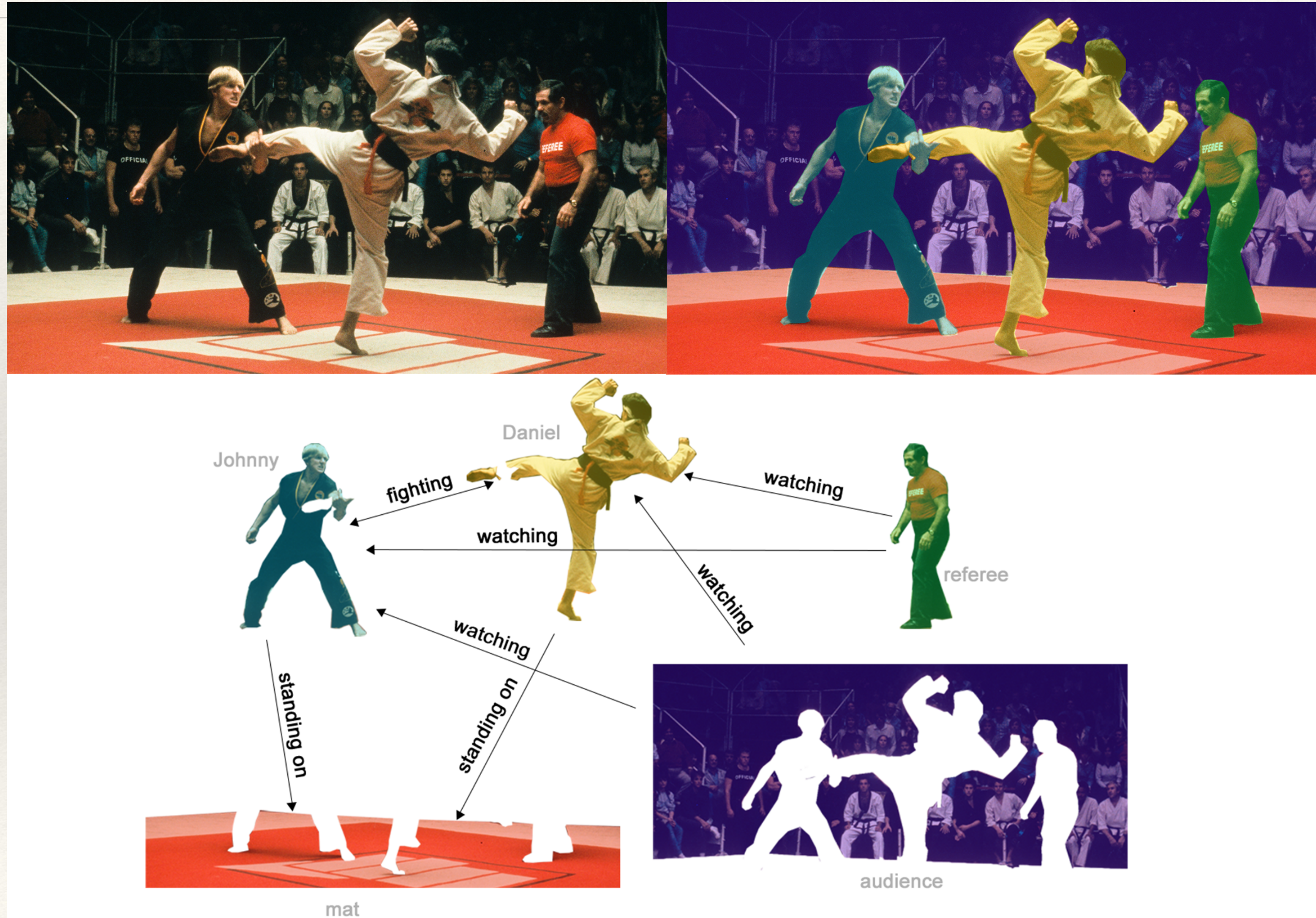
Input: graph with unlabeled nodes

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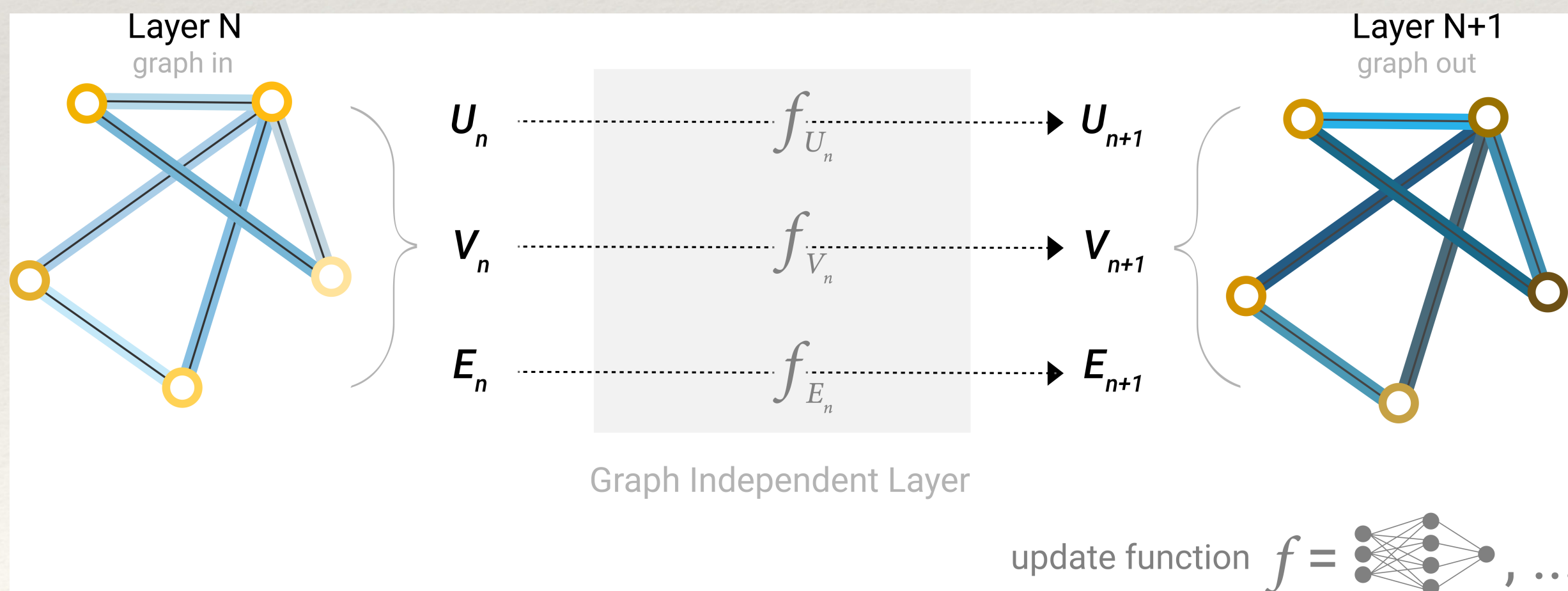
Output: graph node labels

E.g. Edge-level predictions



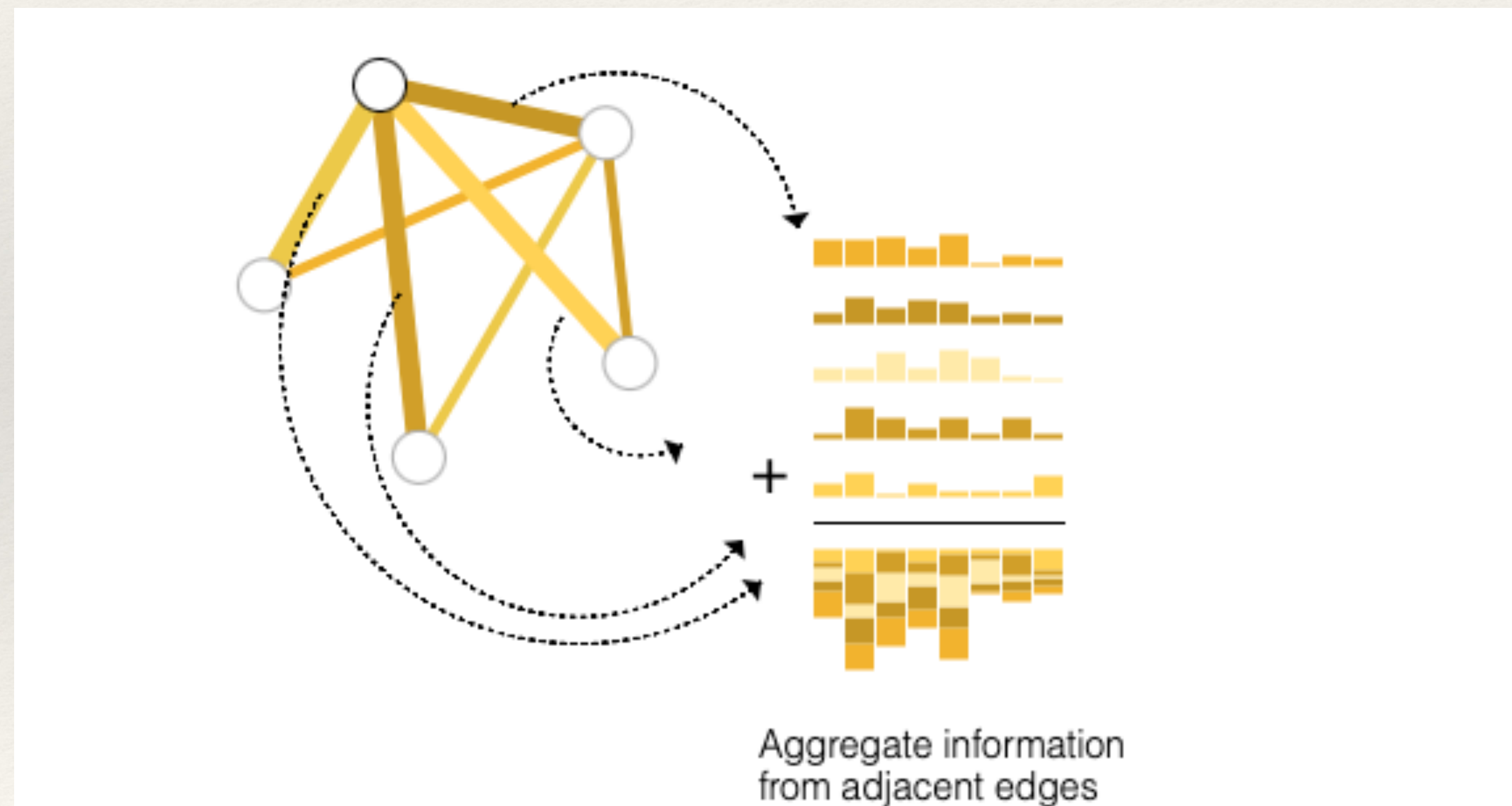
GNN - Message Passing NN(MPNN)

- ❖ Input is a permutation invariant graph(structure preserving) non-regular non-euclidean (with initial vector embeddings for nodes / edges / graph-level. The graph is typically in Adjacency list representation to reduce sparsity
- ❖ Output : Learned vector embeddings for each node / edge / whole graph. The update functions $f_{U_n}, f_{V_n}, f_{E_n}$ are all an MLP! But what is a Message?

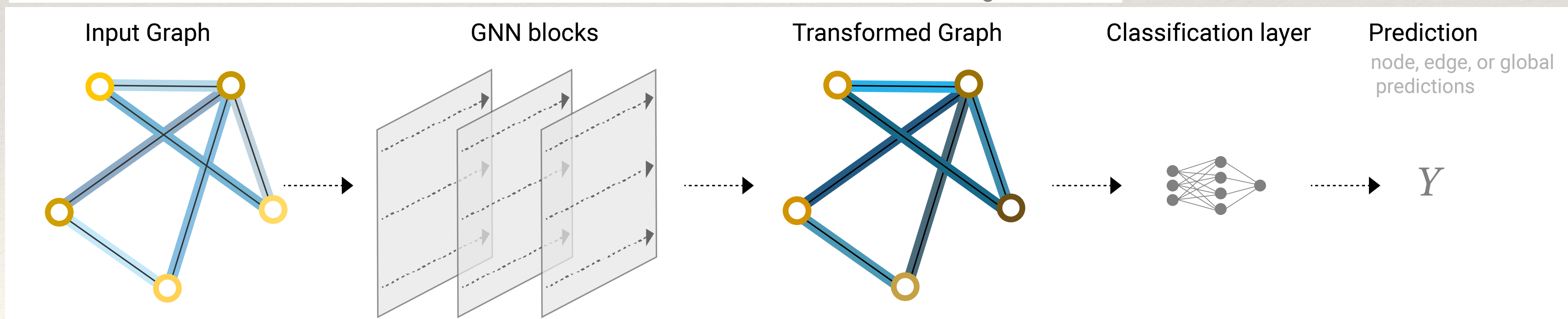
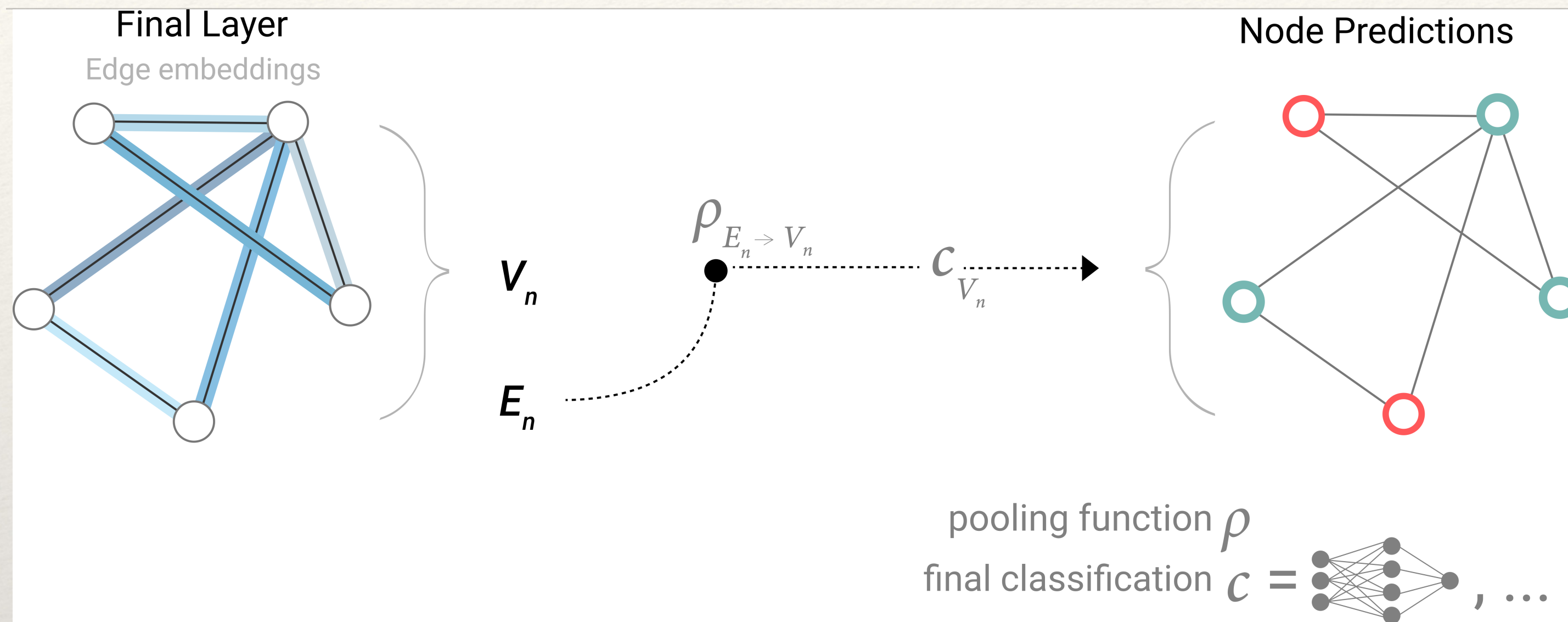


GNN, Message Passing(Signaling)...Pooling, Aggregation

Suppose Information in the graph stored in edges, but no information in nodes, but still need to make predictions on nodes! What can be done? We pool and aggregate information from edges to each node. This is called pooling, message passing or signaling the information and storing it in the nodes(appending to the node's embedding vectors)



GNN - The Flow



Graphon - Limit objects of GNN

- ❖ What happens if Number of nodes $\rightarrow \infty$? Enter Graphons!
- ❖ Graphon : Can be seen as a weighted graph with 'Continuous' node set that is parametrized by a probability space $[0,1]$. More Formally, It is a symmetric, measurable(lebesgue measurable) function $W: [0,1]^2 \rightarrow [0,1]$.
- ❖ For e.g. If x, y are nodes then Edge Weight of a Graphon $W(x, y)$ = Probability of having an edge between x and y
- ❖ Graphon Space $\mathcal{W}_0$ - Set of all Graphons.
- ❖ Graph can be seen as special Graphon.

Graphon - Formal definition

A function W is a **graphon** if it is a **bounded, symmetric, and Lebesgue-measurable function** from the unit square to the unit interval: 

$$W : [0, 1]^2 \rightarrow [0, 1]$$

subject to the following conditions:

- **Symmetry:** $W(x, y) = W(y, x)$ for all $x, y \in [0, 1]$.
- **Measurability:** For any real number a , the set of points $\{(x, y) : W(x, y) \leq a\}$ is a Lebesgue-measurable subset of $[0, 1]^2$.
- **Boundedness:** The function's values are within the interval $[0, 1]$, which can be interpreted as edge weights or connection probabilities. 

Detour - Theorem :Szemerdi Regularity Lemma(weak)

- ❖ **Theorem(Weak Regularity)** : Basically states that as the Number of nodes of a Graph reaches infinity, ANY graph $G(V, E)$ is very close by a small $\epsilon > 0$ to a Graph randomly sampled from a partition of k classes of V , $P = \{V_1, \dots, V_k\}$
- ❖ Here , $k \leq 2^{c/\epsilon^2}$, where c is a constant does not depend on G or ϵ !. Isn't that amazing? The Edge 'density' between Vertices at i, j is given by $d(V_i, V_j) = \frac{e_G(V_i, V_j)}{|V_i||V_j|}$.
- ❖ The expected number of Edges from subsets $U \subset V_i, S \subset V_j$ is given by
$$e_p(U, S) := \sum_{i=1}^k \sum_{j=1}^k \frac{e_G(V_i, V_j)}{|V_i||V_j|} |V_i \cap U| |V_j \cap S|$$
- ❖ $irreg_G(P) = \max_{U, S \subset V} |e_G(U, S) - e_p(U, S)| / |V|^2 \leq \epsilon$ where $e_G(U, S)$ is the number of edges between U and S

What happens to Szemerédi's Lemma for Graphons ?

- ❖ **In Continuous Graphon spaces**, we define what a Graph's Graphon is... Let $I = \{I_1, \dots, I_m\}$ be an equi-partition of $[0,1]$. A Graph $G(V, E)$ with adjacency matrix $A = \{a_{i,j}\} 1 \leq i \leq m, 1 \leq j \leq m$ induces a Graphon $W(x, y) = a_{[xm], [ym]}$. W_G is piecewise constant on I . Thus a Graphon of a Graph can be used for extending the definition of lemma above to Graphon Space
- ❖ Next we define a **Stochastic Block Model(SBM)**. A piecewise constant Graphon like above defined that partitions the Graphon domain $[0,1]$ above with the condition that the number of classes m above is the number of sets in the partition
- ❖ Now we can extend Szemerédi's weak lemma to assert that for some error tolerance $\epsilon > 0$, A graph's Graphon is not that different from a randomly sampled Graphon from SBM of classes $k \leq 2^{c/\epsilon^2}$ for that graph.

Graphon Weak Lemma

- ❖ For precisely formulating it we require a few other definitions

- ❖ First, we define *cut-norm* to be $\| W \| = \sup_{U, S \subset [0,1]} \left| \int_{U \times S} W(x, y) dx dy \right|$. Here U and S are Lebesgue measurable.

- ❖ Next, the *cut-metric*. For $W, V \in W_0$ the Graphon Space, $d(W, V) = \| W - V \|$

- ❖ Then, $S_{[0,1]}$ a space of Measure preserving bijections between the closed intervals $[0,1] \rightarrow [0,1]$. We define $W^\phi = W(\phi(x), \phi(y))$ for each $\phi: A \rightarrow B$ for every $A, B \subset [0,1]$ s.t the sets are measure preserving . i.e. $\mu(A) = \mu(\phi(A))$. Note that there could be null-sets in the interval with just a single element and co-null set with complement of the union of null-sets. For null-set the measure is 0 and we take W to be 0.

Graphon Weak Lemma contd...

- ❖ Now the irregularity can be defined in Graphon space as $\delta(W, W^\phi) = \inf_{\phi \in S_{[0,1]}} \|W - W^\phi\|$
where $\|W - W^\phi\| = \sup_{S, T \subseteq [0,1]} \left| \int_{S \times T} (W(x, y) - W(\phi(x), \phi(y))) dx dy \right|$. Thus $\delta(W, W^\phi) \leq \epsilon$
- ❖ **Theorem (Graphon weak regularity)** : For every graphon $W \in W_0$ and $\epsilon > 0$ there exists a step Graphon $W' \in W_0$ (Stochastic Block Model) w.r.t to partition of utmost $\left\lceil 2^{c/\epsilon^2} \right\rceil$ s.t $\delta(W, W^\phi) \leq \epsilon$ for some universal constant c.
- ❖ It can be seen that the SBM is dense in W_0 the Graphon Space and with the induced measure it can be shown as Compact as well(proved in \cite{1})

So, what does that mean?

- ❖ This means for Very large graphs we can work with the Graphon induced by the Partition class with finite set of classes way less than the number of nodes in G and We can measure Similarities between Graphs that are very large , study isomorphisms etc. Transfer Learnings(the learnt embeddings for node / edge / Graph level) from One large Graph to another for Similar problems , if they seem close. This saves a lot of computationally intensive retraining!

Graphon-Signals and Lipschitz continuity

- ❖ Similar metric and definitions can be extended to the values of the Learnt embeddings in Graphon-Signalling space. Interested Reader can refer to Section 4. Graphon-Signal Analysis of MPNN's in the paper "A Graphon-signal analysis of graph neural networks" by Ron Levie. This is the most important part proving that the learnt values of the embeddings by the update function are going to be very similar to a random sampling from a SBM of Graphon-Signal induced by the MPNN transformations
- ❖ Lipschitz continuity ensures that small perturbations in input , say due to aggregation/ pooling , results in very small changes in learnt embedding values. This is an important requirement for any Model.

SBM $\rightarrow$ A Topological space

- ❖ Why would we require a possible Topological Model for SBMs?
- ❖ A: It is possible that while training two different GNN for problem A and B with very large number of nodes, the SBM for both are possibly very similar. i.e the Layer 1...Layer K transformations of the embeddings are pointing to very similar SBM values (w.r.t to Signaling) after possibly future 3 months of training With close values. Months of training for B could be avoided if somehow we can detect that the final set of embeddings that are going to be learnt for the new problem is very close to an already trained GNN for A which the learnt embeddings ARE already available. How do we even recognize that?
- ❖ Next, the embeddings are very very problem specific. The input and output are Model and notation and pre-assigned Sample distribution probability specific. How do we compare A and B in two completely different problem domains?

The Topological Space proposal

- ❖ Before laying out the Topological extensions for SBM, First off, the Number of partitions at each step that are being produced for induced Graphon for GNN A and B can be compared . Because of the regularity Lemma for Graphon-Signals for Graphs that are $n \geq 2^{\lceil \frac{9c}{4\epsilon^2} \rceil}$ (yes different than for Graphons) the cut-distance between signals(learnt values) is less-than ϵ . This can be done today without Topological characterization.
- ❖ That said, The proposal is as follows. Suppose we have a map(linear or non-linear) from the SBM Interval I for the graphon-signal to a n -sphere, n being the number of classes of the SBM such that distinct subsets map to distinct regions on the Surface of the n-Sphere with the measure space intact, Then we will have a continuous map. The map can be a isometry(replacing lebesgue measure) . If it is a bijective , continuous map then it becomes a homeomorphism. In this case the Surface area for a n-ball for e.g is given by $S_{n-1} = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})}$ where γ is the Gamma function. This surface Area(Or Hausdorff Measure) is 1 as the probability values are 1.

The n-Sphere mapping

- ❖ Therefore, we could do two things. A) fix the surface area to be 1 and find the dimension. B) Fix the dimension and find the radius.

❖ For, which n the Surface area is 1? To solve this, We must have $1 = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})}$, Since, this is has a

Transcendental solution, No analytical computation can yield the solution. Numerically computing this gives $n \approx 17.768281888226918$. The Area then is $A_{17.768281888226918} = 1.000000000000000001 \approx 1$, close-enough!

- ❖ Next, for e,g We fix the dimension to be an integer for e.g 3-Sphere and get the area to be 1 . i.e $4\pi r^2 = 1 \implies r = \frac{1}{2\sqrt{\pi}}$. Next , I did use AI help to find the mapping I want(I hope this is ok!).

SBM $\rightarrow$ n-Sphere map

2. Target Space: The Normalized Sphere

$$S^N = \{z \in \mathbb{R}^{N+1} : \|z\| = 1\}, \quad \sigma_N(S^N) = 1.$$

When $N \approx 17.768$, the unit sphere has surface area 1. For visualization, we often use S^2 .

Goal

Find a measurable map

$$\Phi : [0, 1] \longrightarrow S^N$$

that converts SBM blocks into distinct spherical regions of proportional area.

SBM $\rightarrow$ n-Sphere map

3. Construction of the Map Φ

Step 1. Centers:

Choose $v_1, \dots, v_n \in S^N$ well-separated (e.g. simplex vertices).

Step 2. Caps:

Define

$$C_i = \{z \in S^N : \langle z, v_i \rangle \geq \cos r_i\},$$

where the radius r_i satisfies

$$\sigma_N(C_i) = p_i \quad \Leftrightarrow \quad p_i = \frac{\int_0^{r_i} \sin^{N-1} \theta \, d\theta}{\int_0^\pi \sin^{N-1} \theta \, d\theta} = I_{\sin^2(r_i)}\left(\frac{N}{2}, \frac{1}{2}\right).$$

Step 3. Parametrization:

Define a measure-preserving map $\phi_i : I_i \rightarrow C_i$.

Step 4. Global map:

$$\Phi(x) = \phi_i(x) \quad \text{if } x \in I_i.$$

SBM $\rightarrow$ n-Sphere map

4. Compact Expression

$$F_N(r) = \frac{\int_0^r \sin^{N-1} \theta \, d\theta}{\int_0^\pi \sin^{N-1} \theta \, d\theta}, \quad r_i = F_N^{-1}(p_i).$$

Then

$$\Phi(x) = R(v_i, \Theta(x)), \quad x \in I_i,$$

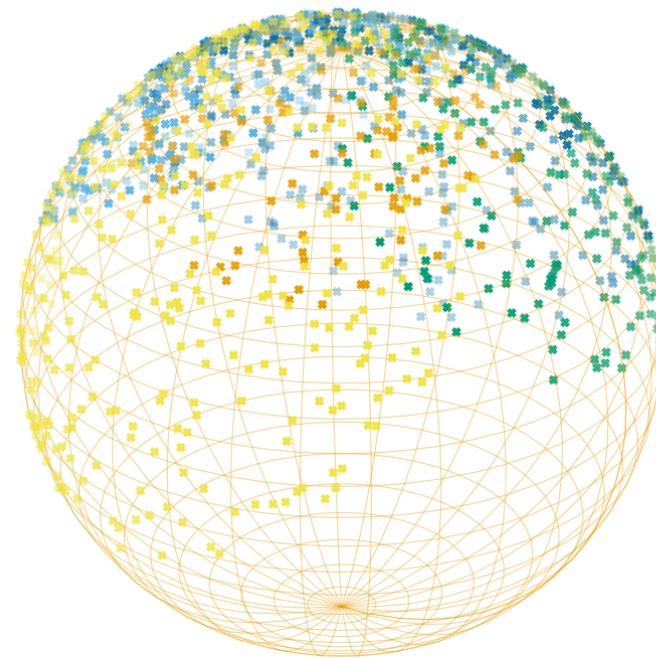
where $\Theta(x)$ satisfies

$$F_N(\Theta(x)) = \frac{x - \inf I_i}{p_i},$$

and $R(v_i, \Theta)$ rotates the north pole to v_i with polar angle Θ .

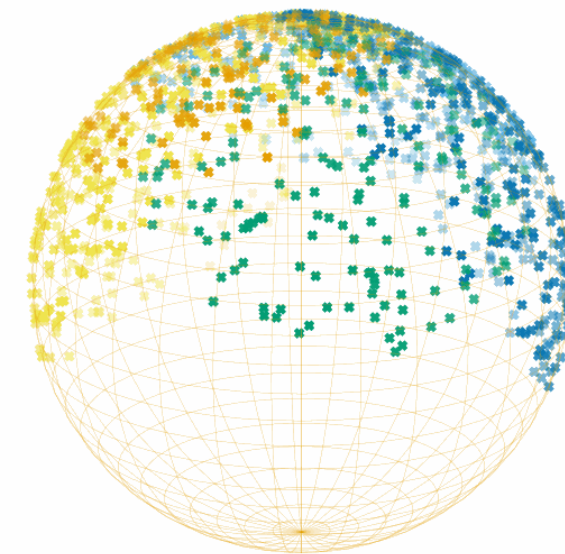
SBM $\rightarrow$ n-Sphere map

Illustration: 5-class SBM blocks mapped to spherical caps on S^2
(areas proportional to block sizes)



- class 1 ($p=0.10$)
- class 2 ($p=0.20$)
- class 3 ($p=0.15$)
- class 4 ($p=0.30$)
- class 5 ($p=0.25$)

5-class SBM mapped to spherical caps (rotating view)



- class 1 ($p=0.10$)
- class 2 ($p=0.20$)
- class 3 ($p=0.15$)
- class 4 ($p=0.30$)
- class 5 ($p=0.25$)

Conclusion & Further Research

- ❖ The illustration is on 3-Sphere. But possibly the points cluster around for larger number of embeddings. For e.g LLMs use 700-800 values for each word.
- ❖ So what have we achieved? For Model Developers they could publish the Image along with their model and a New problem class GNN could refer to existing images GIFs to look at existing classes for their GNN and possibly there may be a chance then the only pending work is to Re-map the points to their embedding space that corresponds to the new problem!
- ❖ Further Research: 1) We also have to consider other Spaces besides n -Sphere. The Geometry of the non-linearity would dictate possibly other Manifolds.

References

- ❖ 1) Franzosa, Adams : *Introduction to Topology*
- ❖ 2) *Algebraic Topology*, Hatcher.
- ❖ 3) *Fine-grained Expressivity of Graph Neural Networks* , Ron Levie et.al. NIPS '23 proceedings
- ❖ 4) *Orthogonal bases for Equivariant Graph Learning with Provable k-WL Expressive power*, Maggie Cheng, Jia He
- ❖ 5) *Generalization, Expressivity and Universality of Graph Neural Networks on Attributed Graphs*
- ❖ 6) *A graph-on Signal Analysis of Graph Neural Networks* NIPS '23
- ❖ 7) <https://distill.pub/2021/gnn-intro/> for Visualizations

Thank You!